MLOps Assignment 2

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1. Which run had the lowest latency, and why might that be?

- run_batch4 had the lowest latency, at 0.104 seconds.
- run_batch8 took 0.116 seconds, and run_batch1 was the slowest at 0.358 seconds.
- Batch processing allows the model to process multiple sentences together, which can reduce the overhead of making separate inference calls.
- run_batch1 processes the sentences with a batch size of 1, while batch sizes 4 and 8 process multiple sentences together.
- In my experiment, batch size 4 gave the lowest measured latency. The fastest batch size can depend on the model, hardware, and workload.

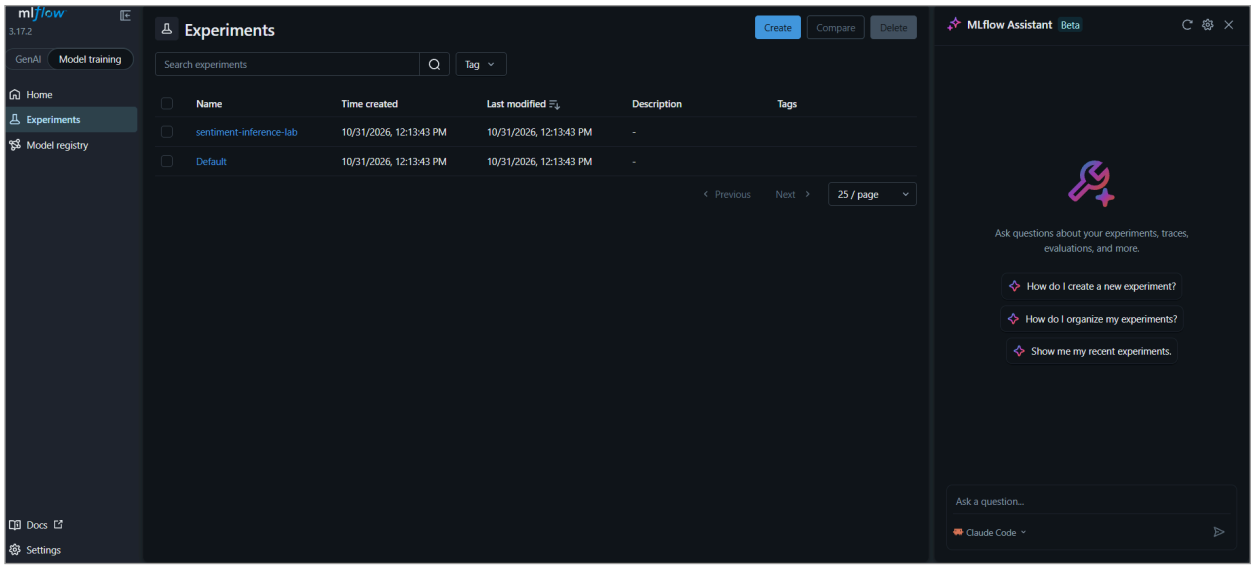
2. Did accuracy change between runs? Why or why not?

- No, accuracy remained 1.0 (100%) in all three runs.
- Every run used the same pre-trained DistilBERT sentiment-analysis model.
- All runs were tested on the same 10 sentences.
- Changing the batch size affected the processing time, but it did not change the predicted sentiment labels in these experiments.
- Therefore, all three runs achieved the same accuracy.

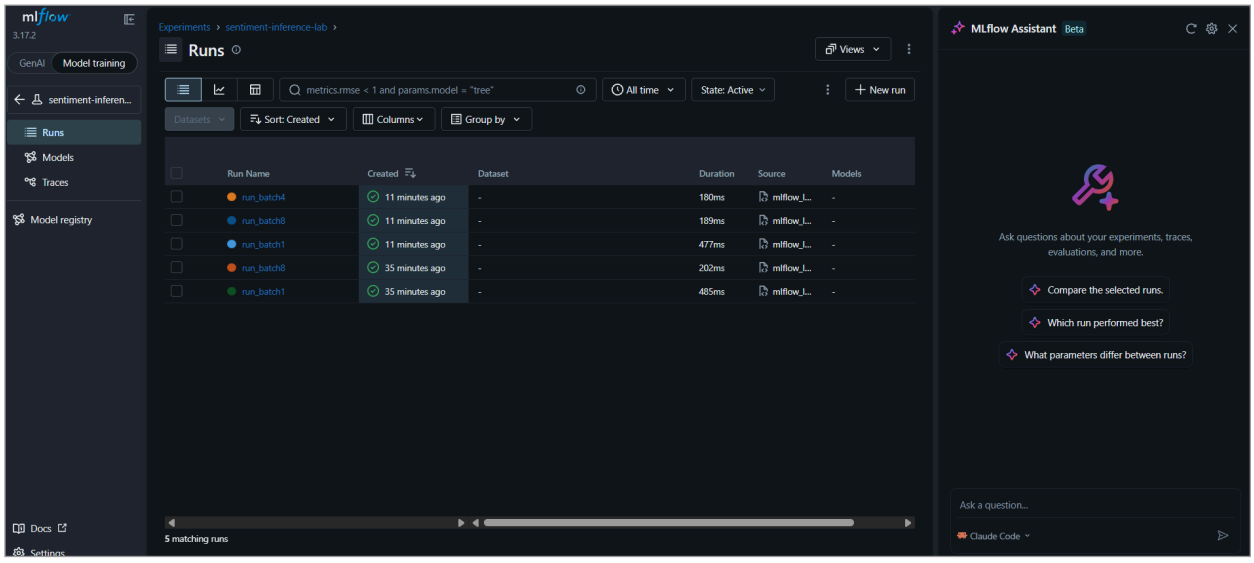
3. Name two benefits of using MLflow here instead of printing results in the terminal.

- Experiment tracking and record-keeping: MLflow records the parameters, metrics, and artifacts such as predictions.csv for each experiment run, making the results easier to review later.
- Easy comparison: The MLflow UI allows experiment runs to be viewed and compared using their parameters and metrics, making performance differences easier to identify than reading terminal output.

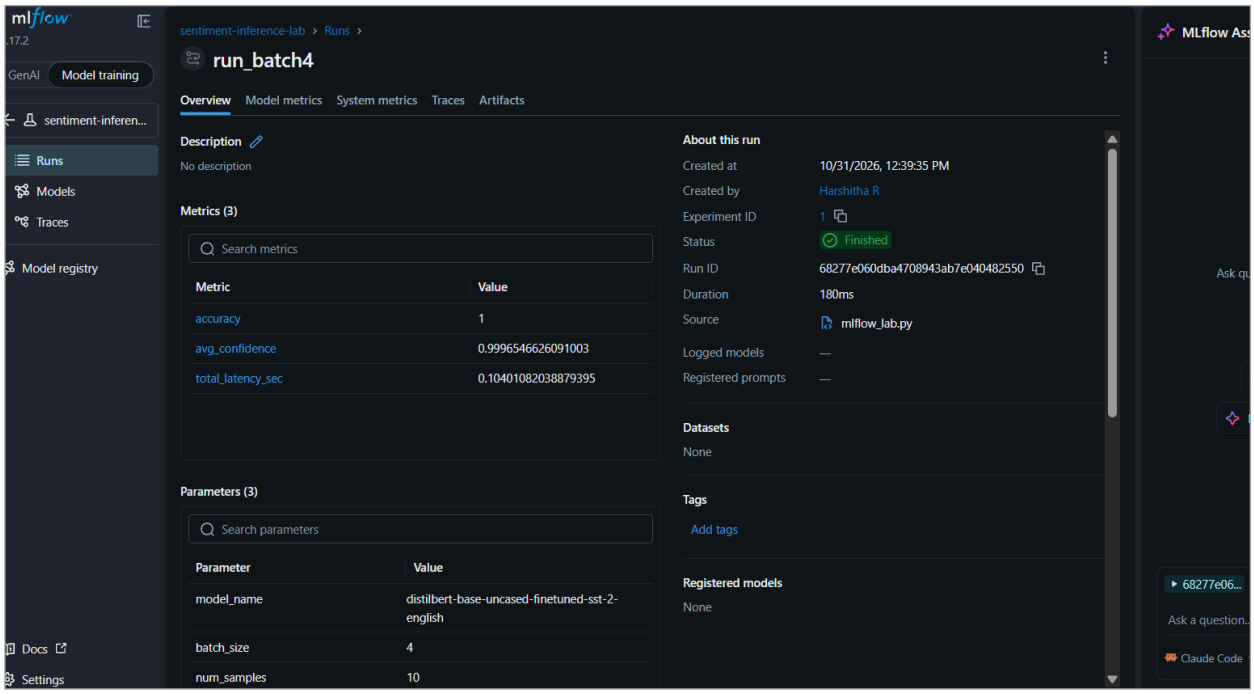
Screenshot of the MLflow experiments list:



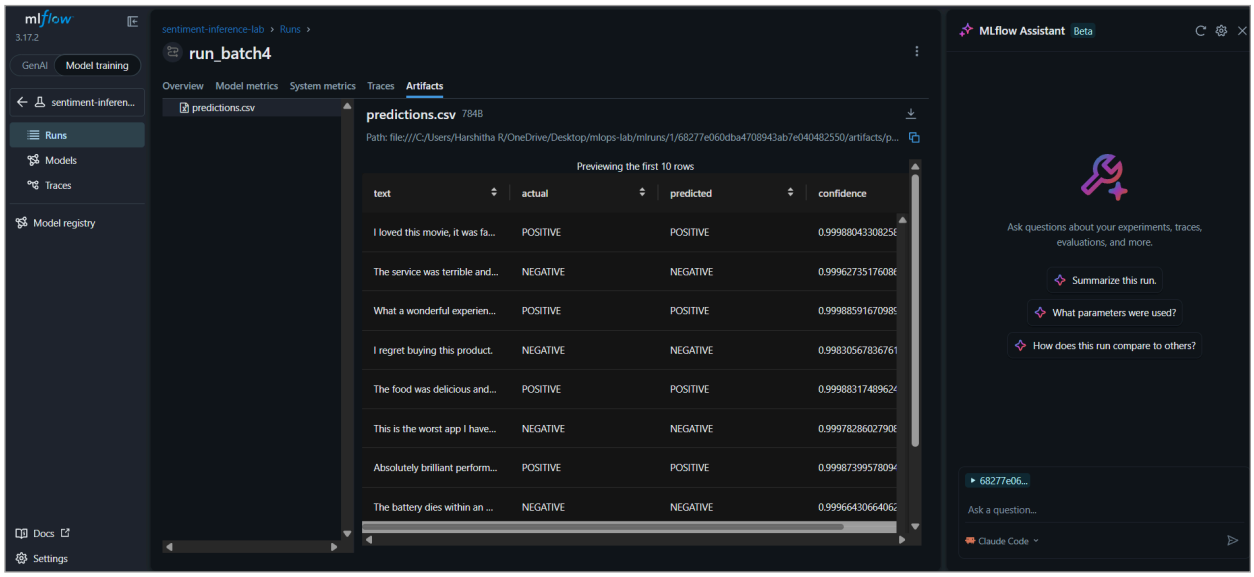
Screenshot of the MLflow runs table:



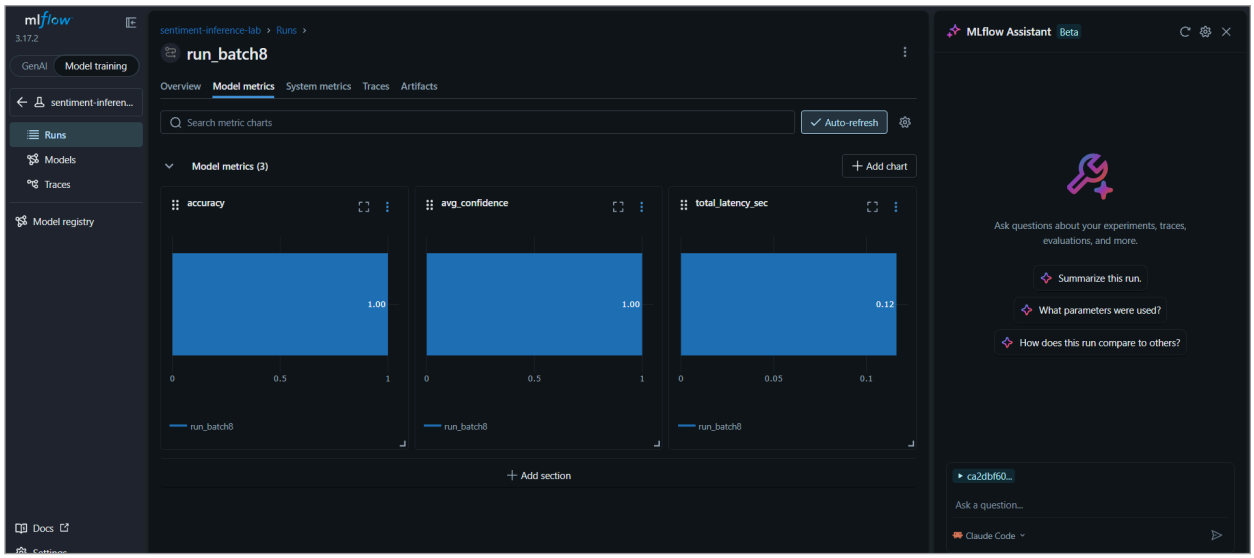
run_batch4 - Overview:



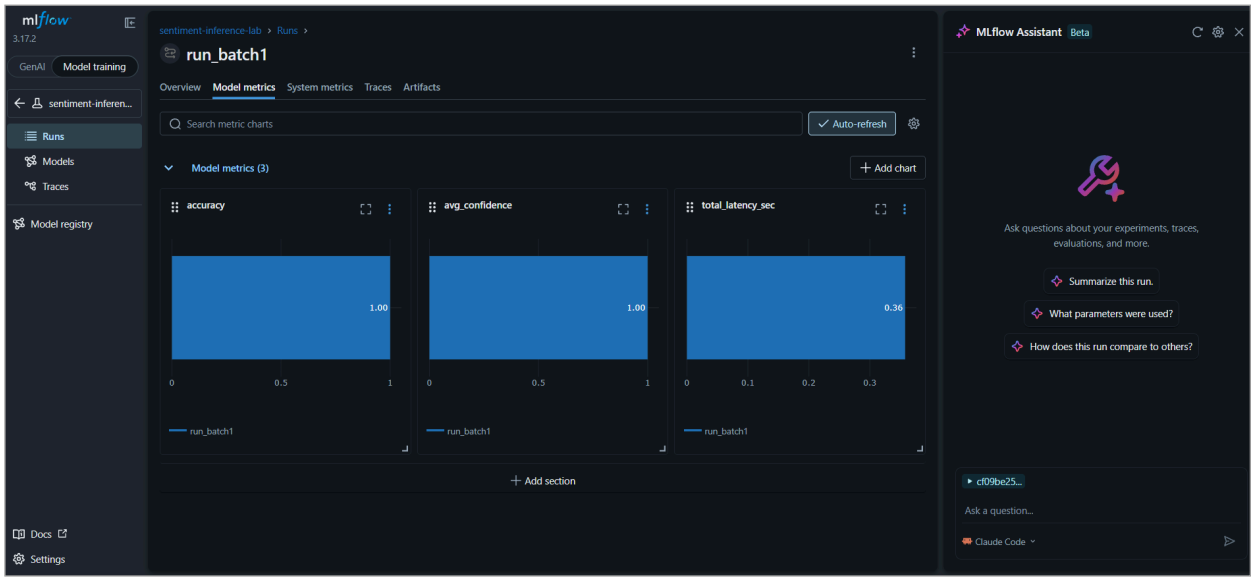
run_batch4 - Artifacts (predictions.csv):



run_batch8 - Model metrics:



run_batch1 - Model metrics:



Execution:

```
(mlops-env) C:\Users\Harshitha R\Desktop\mlops-lab>python mlflow_lab.py
Loading weights: 100%|████████████████████| 104/104 [00:00<00:00, 471.42it/s]
run_batch1 -> accuracy: 1.0 latency: 0.358
Loading weights: 100%|████████████████████| 104/104 [00:00<00:00, 482.61it/s]
run_batch8 -> accuracy: 1.0 latency: 0.116
Loading weights: 100%|████████████████████| 104/104 [00:00<00:00, 800.95it/s]
run_batch4 -> accuracy: 1.0 latency: 0.104

(mlops-env) C:\Users\Harshitha R\Desktop\mlops-lab>findstr /n "tracking_uri
set_experiment start_run" mlflow_lab.py
22:mlflow.set_tracking_uri("sqlite:///mlflow.db")
23:mlflow.set_experiment("sentiment-inference-lab")
29:    with mlflow.start_run(run_name=run_name):
```